Illustrative Mathematics 高中（代数1 / 几何 / 代数2）大纲与海报计划

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海报粒度：和 IM 6–8 一样，每个 Section 一张。全套 102 个 Section（代数1 37、几何 32、代数2 33），不超过约 250 张，所以按 Section 出海报。每个单元最后的「Let’s Put It to Work（综合运用）」部分官方没有列 Section Goals，大纲里改为摘录该部分的说明。标注 (Optional) 的课是官方的选学课。

本地化：海报只用公制单位和人民币；为说明而举的数字和数据都标了「示例」，示意图标「示意」；涉及动手操作（圆规、剪刀、镜子、运动测心率等）的海报加了安全提示，涉及的 Section 下面有标注。

来源

- Algebra 1：课程首页 · Scope and Sequence · 单元页如 Unit 1
- Geometry：课程首页 · Scope and Sequence · 单元页如 Unit 1
- Algebra 2：课程首页 · Scope and Sequence · 单元页如 Unit 1
- 总页：IM 9–12 Math（accessim.org） · IM 9–12 Math v.360 介绍 · 授权说明

代数1（Algebra 1）· 8 个单元 · 37 个 Section

第1单元 单变量统计（Unit 1: One-Variable Statistics）

单元页：Algebra 1 Unit 1

A部分 认识彼此（Section A: Getting to Know You）

- 学习目标：Create and interpret data displays such as dot plots, histograms, and box plots.
- 课：1.1 Getting to Know You；1.2 Data Representations (Optional)；1.3 A Gallery of Data
- 海报要点：用点图、直方图、箱线图展示和解读班级调查数据
- 海报：`a1-u1a-认识彼此`「数据怎样展示？」
- 本地化 / 安全：海报只用公制单位，调查数据是示例。

B部分 分布的形状（Section B: Distribution Shapes）

- 学习目标：Describe the shape of a distribution, including a measure of center and a measure of variability.
- 课：1.4 The Shape of Distributions；1.5 Calculating Measures of Center and Variability (Optional)
- 海报要点：对称、偏态、钟形、均匀、双峰；用中心和离散程度描述分布
- 海报：`a1-u1b-分布的形状`「分布是什么形状？」

C部分 使用电子表格（Section C: How to Use Spreadsheets）

- 学习目标：Use a spreadsheet as a tool for calculating values quickly.
- 课：1.6 Mystery Computations；1.7 Spreadsheet Computations (Optional)；1.8 Spreadsheet Shortcuts (Optional)
- 海报要点：单元格公式、填充柄，快速批量计算
- 海报：`a1-u1c-使用电子表格`「电子表格怎样算？」

D部分 处理数据（Section D: Manipulating Data）

- 学习目标：Describe the effect of outliers on a distribution of data.
- 学习目标：Recognize standard deviation as a measure of variability.
- 课：1.9 Technological Graphing；1.10 The Effect of Extremes；1.11 Comparing and Contrasting Data Distributions；1.12 Standard Deviation；1.13 More Standard Deviation；1.14 Outliers；1.15 Comparing Data Sets
- 海报要点：极端值对平均数和中位数的影响；标准差；离群值判断（1.5 × IQR）；比较数据集
- 海报：`a1-u1d-处理数据`「离群值有什么影响？」

E部分 综合运用：分析数据（Section E: Let's Put It to Work）

- 说明（官方未列 Section Goals）：The final section of this unit is a lesson where students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：1.16 Analyzing Data
- 海报要点：提出统计问题，收集、展示、分析数据并汇报
- 海报：`a1-u1e-综合运用分析数据`「怎样分析一组数据？」

第2单元 线性方程和方程组（Unit 2: Linear Equations and Systems）

单元页：Algebra 1 Unit 2

A部分 用方程建模（Section A: Writing and Modeling with Equations）

- 学习目标：Interpret solutions to equations in one variable and in two variables.
- 学习目标：Recognize that the graph of a linear equation in two variables represents all pairs of values that are solutions to the equation.
- 学习目标：Write equations with numbers and variables to describe relationships and constraints.
- 课：2.1 Planning a Party；2.2 Writing Equations to Model Relationships (Part 1) (Optional)；2.3 Writing Equations to Model Relationships (Part 2)；2.4 Equations and Their Solutions (Optional)；2.5 Equations and Their Graphs (Optional)
- 海报要点：用含变量的方程描述关系和限制条件；方程的解；二元方程的图像是所有解
- 海报：`a1-u2a-用方程建模`「方程怎样描述情境？」
- 本地化 / 安全：聚会预算的金额用人民币（元），数字是示例。

B部分 变形方程与结构（Section B: Manipulating Equations and Understanding Their Structure）

- 学习目标：Determine the slope and vertical intercept of the graphs of linear equations by making use of structure or by rearranging the equations.
- 学习目标：Rearrange multi-variable equations to highlight a particular quantity.
- 学习目标：Recognize that “equivalent equations” are equations that have exactly the same solutions, and that multiple equivalent equations can represent the same relationship.
- 课：2.6 Equivalent Equations；2.7 Explaining Steps for Rewriting Equations；2.8 Which Variable to Solve for? (Part 1)；2.9 Which Variable to Solve for? (Part 2)；2.10 Connecting Equations to Graphs (Part 1)；2.11 Connecting Equations to Graphs (Part 2)
- 海报要点：等价方程；解释变形步骤；按需要解出某个变量；由方程读斜率和截距
- 海报：`a1-u2b-变形方程与结构`「怎样变形方程？」

C部分 二元一次方程组（Section C: Systems of Linear Equations in Two Variables）

- 学习目标：Determine whether a system of equations will have 0, 1, or infinitely many solutions by analyzing their structure or by graphing.
- 学习目标：Use elimination or substitution to create one or more equivalent systems to help solve the original system.
- 课：2.12 Writing and Graphing Systems of Linear Equations；2.13 Solving Systems by Substitution；2.14 Solving Systems by Elimination (Part 1)；2.15 Solving Systems by Elimination (Part 2)；2.16 Solving Systems by Elimination (Part 3)；2.17 Systems of Linear Equations and Their Solutions
- 海报要点：画图找交点；代入法、消元法；无解、唯一解、无数解
- 海报：`a1-u2c-二元一次方程组`「方程组怎样解？」

D部分 综合运用（Section D: Let's Put It to Work）

- 说明（官方未列 Section Goals）：In the final section, students have the opportunity to apply their thinking from throughout the unit. Because this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：2.18 Asking about Solving Systems (Optional)；2.19 Linear Patterns (Optional)
- 海报要点：就解方程组提问；线性规律

- 海报：`a1-u2d-综合运用`「怎样用好方程组？」

第3单元 双变量统计（Unit 3: Two-Variable Statistics）

单元页：Algebra 1 Unit 3

A部分 双向表（Section A: Two-Way Tables）

- 学习目标：Create relative frequency tables from information given in a two-way table or about a situation.
- 学习目标：Inspect patterns in relative frequency tables and two-way tables to determine if there is a possible association between two variables of interest.
- 课：3.1 Two-Way Tables；3.2 Relative Frequency Tables；3.3 Associations in Categorical Data
- 海报要点：双向频数表、相对频率表；从表中判断两个分类变量是否有关联
- 海报：`a1-u3a-双向表`「两个分类变量有关吗？」

B部分 散点图（Section B: Scatter Plots）

- 学习目标：Comprehend the connection between residuals, variability, and whether or not using a linear model is appropriate.
- 学习目标：Interpret the rate of change and vertical intercept for a linear model in the context of a situation.
- 课：3.4 Linear Models；3.5 Fitting Lines；3.6 Residuals
- 海报要点：拟合直线；斜率和截距在情境中的意义；残差
- 海报：`a1-u3b-散点图`「怎样画拟合直线？」

C部分 相关系数（Section C: Correlation Coefficients）

- 学习目标：Describe the strength and sign of the relationship between variables based on the correlation coefficient.
- 学习目标：Investigate the relationship between two variables to analyze whether or not the relationship is causal.
- 课：3.7 The Correlation Coefficient；3.8 Using the Correlation Coefficient；3.9 Causal Relationships
- 海报要点：r 的符号和大小表示关联方向和强弱；相关不等于因果
- 海报：`a1-u3c-相关系数`「相关系数说明什么？」

D部分 综合运用：化石与旗杆（Section D: Let's Put It to Work）

- 说明（官方未列 Section Goals）：In the final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：3.10 Fossils and Flags (Optional)
- 海报要点：用线性模型由骨长估计身高
- 海报：`a1-u3d-综合运用化石与旗杆`「骨长能估计身高吗？」
- 本地化 / 安全：骨长和身高用厘米，数据是示例。

第4单元 线性不等式和不等式组（Unit 4: Linear Inequalities and Systems）

单元页：Algebra 1 Unit 4

A部分 一元一次不等式（Section A: Linear Inequalities in One Variable）

- 学习目标：Use a related equation to solve an inequality in one variable.
- 学习目标：Write and solve inequalities in one variable to represent the constraints in situations and to solve problems.
- 课：4.1 Representing Situations with Inequalities；4.2 Solutions to Inequalities in One Variable；4.3 Writing and Solving Inequalities in One Variable
- 海报要点：用不等式表示限制；借助相应方程解不等式，数轴表示解集
- 海报：`a1-u4a-一元一次不等式`「不等式怎样解？」
- 本地化 / 安全：情境里的金额用人民币，数字是示例。

B部分 二元一次不等式（Section B: Linear Inequalities in Two Variables）

- 学习目标：Understand that a constraint on two variables can be represented by an inequality, a graph (a half-plane), and a verbal description.
- 学习目标：Write inequalities in two variables to represent the constraints in a situation, and use technology to graph the solution set to answer questions about the situation.
- 课：4.4 Graphing Linear Inequalities in Two Variables (Part 1)；4.5 Graphing Linear Inequalities in Two Variables (Part 2)；4.6 Solving Problems with Inequalities in Two Variables
- 海报要点：解集是半平面；边界线实线或虚线；检验点
- 海报：`a1-u4b-二元一次不等式`「二元不等式怎样画？」

C部分 二元一次不等式组（Section C: Systems of Linear Inequalities in Two Variables）

- 学习目标：Given a system of inequalities and their graphs, explain how to tell if a pair of values is a solution to the system.
- 学习目标：Understand that the solution set of a system of inequalities in two variables is composed of any pair of values that make both inequalities true, and that it is represented graphically by the region where the graphs overlap.
- 课：4.7 Solutions to Systems of Linear Inequalities in Two Variables；4.8 Solving Problems with Systems of Linear Inequalities in Two Variables；4.9 Modeling with Systems of Inequalities in Two Variables
- 海报要点：解集是各半平面的公共区域；判断某点是否为解；建模
- 海报：`a1-u4c-二元一次不等式组`「不等式组的解在哪？」

第5单元 函数（Unit 5: Functions）

单元页：Algebra 1 Unit 5

A部分 函数及其表示（Section A: Functions and Their Representations）

- 学习目标：Sketch a graph of a function given statements in function notation.
- 学习目标：Understand that a relationship between two quantities is a function if there is only one possible output for each input.
- 学习目标：Write equations that represent rules using function notation.
- 课：5.1 Describing and Graphing Situations；5.2 Function Notation；5.3 Interpreting and Using Function Notation；5.4 Using Function Notation to Describe Rules (Part 1)；5.5 Using Function Notation to Describe Rules (Part 2)
- 海报要点：每个输入只有一个输出；函数记号 f(x)；由描述画图

- 海报：`a1-u5a-函数及其表示`「什么是函数？」

B部分 分析和绘制函数图像（Section B: Analyzing and Creating Graphs of Functions）

- 学习目标：Given a graph of a function, estimate or calculate the average rate of change over a specified interval.
- 学习目标：Interpret key features of a graph—the intercepts, maximums, minimums, and intervals when the function is increasing or decreasing—in terms of a situation.
- 学习目标：Interpret statements about two or more functions written in function notation.
- 课：5.6 Features of Graphs；5.7 Using Graphs to Find Average Rate of Change；5.8 Interpreting and Creating Graphs；5.9 Comparing Graphs
- 海报要点：截距、最大最小值、增减区间；平均变化率
- 海报：`a1-u5b-分析和绘制函数图像`「函数图像怎样读？」

C部分 细看输入和输出（Section C: A Closer Look at Inputs and Outputs）

- 学习目标：Interpret an absolute value function described in words or in function notation, and create a table of values and a graph to represent the function.
- 学习目标：Interpret the graph of a piecewise function or its rules given in function notation, and explain the rules (orally and in writing) in terms of a situation.
- 学习目标：Understand that the domain of a function is the set of all possible inputs and the range is the set of all possible outputs.
- 课：5.10 Domain and Range (Part 1)；5.11 Domain and Range (Part 2)；5.12 Piecewise Functions；5.13 Absolute Value Functions (Part 1)；5.14 Absolute Value Functions (Part 2)
- 海报要点：定义域和值域；分段函数；绝对值函数
- 海报：`a1-u5c-细看输入和输出`「定义域和值域是？」

D部分 反函数（Section D: Inverse Functions）

- 学习目标：Find the inverse of a linear function by solving an equation for the input variable.
- 学习目标：Write a linear function and an inverse function to model data and solve problems.
- 课：5.15 Inverse Functions；5.16 Finding and Interpreting Inverse Functions；5.17 Writing Inverse Functions to Solve Problems
- 海报要点：交换输入输出；解出输入变量得到反函数
- 海报：`a1-u5d-反函数`「反函数怎样求？」

E部分 综合运用：电池电量（Section E: Let's Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：5.18 Using Functions to Model Battery Power
- 海报要点：用函数为电池电量建模
- 海报：`a1-u5e-综合运用电池电量`「电池能用多久？」

第6单元 指数函数入门（Unit 6: Introduction to Exponential Functions）

单元页：Algebra 1 Unit 6

A部分 观察增长（Section A: Looking at Growth）

- 学习目标：Compare linear and exponential relationships by performing calculations.
- 学习目标：Describe patterns in tables that represent linear and exponential relationships.
- 课：6.1 Growing and Growing；6.2 Patterns of Growth
- 海报要点：线性每次加同一个数，指数每次乘同一个数
- 海报：`a1-u6a-观察增长`「加还是乘？」

B部分 一种新的关系（Section B: A New Kind of Relationship）

- 学习目标：Interpret a negative exponent in equations that represent exponential growth or decay.
- 学习目标：Write an equation of the form $y=a \cdot b^x$ to represent a quantity a that changes by a growth factor b.
- 学习目标：Write and graph an equation that represents exponential decay to solve problems.
- 课：6.3 Representing Exponential Growth；6.4 Representing Exponential Decay；6.5 Understanding Decay；6.6 Analyzing Graphs；6.7 Using Negative Exponents
- 海报要点：$y = a \cdot b^x$；增长因子、衰减因子；负指数的意义
- 海报：`a1-u6b-一种新的关系`「什么是指数关系？」

C部分 指数函数（Section C: Exponential Functions）

- 学习目标：Describe the effect of changing a and b on a graph that represents $f(x) = a \cdot b^x$.
- 学习目标：Use function notation to write equations that represent exponential relationships.
- 课：6.8 Exponential Situations as Functions；6.9 Interpreting Exponential Functions；6.10 Looking at Rates of Change；6.11 Modeling Exponential Behavior；6.12 Reasoning about Exponential Graphs (Part 1)；6.13 Reasoning about Exponential Graphs (Part 2) (Optional)
- 海报要点：$f(x) = a \cdot b^x$；a、b 对图像的影响；平均变化率
- 海报：`a1-u6c-指数函数`「a、b 怎样影响图像？」

D部分 百分比增长与衰减（Section D: Percent Growth and Decay）

- 学习目标：Calculate the result of repeated percent increase for the same initial balance and interest rate, but compounded at different intervals.
- 学习目标：Justify why applying a percent increase p, n times, is not equivalent to applying the percent np.
- 课：6.14 Recalling Percent Change (Optional)；6.15 Functions Involving Percent Change；6.16 Compounding Interest；6.17 Different Compounding Intervals；6.18 Expressed in Different Ways
- 海报要点：复利；不同计息周期；连续增长 p% 共 n 次不等于增长 np%
- 海报：`a1-u6d-百分比增长与衰减`「复利怎样计算？」
- 本地化 / 安全：存款和利息用人民币，利率是示例，不代表真实银行利率。

E部分 比较线性和指数函数（Section E: Comparing Linear and Exponential Functions）

- 学习目标：Use rates of change, and show that, for any equal intervals of the independent variable, an exponential function always increases or decreases by an equal factor.
- 学习目标：Use tables, calculations, and graphs to compare growth rates of linear and exponential functions.
- 课：6.19 Which One Changes Faster?；6.20 Changes over Equal Intervals
- 海报要点：相等间隔：线性加相同的量，指数乘相同的倍数；指数终将超过线性
- 海报：`a1-u6e-比较线性和指数函数`「谁增长得更快？」

F部分 综合运用：预测人口（Section F: Let's Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：6.21 Predicting Populations
- 海报要点：用线性或指数模型预测人口
- 海报：`a1-u6f-综合运用预测人口`「人口会怎样变化？」

第7单元 二次函数入门（Unit 7: Introduction to Quadratic Functions）

单元页：Algebra 1 Unit 7

A部分 另一种变化（Section A: A Different Kind of Change）

- 学习目标：Comprehend that a “quadratic relationship” can be expressed with a squared term.
- 学习目标：Determine and explain whether a visual pattern represents a linear, exponential, or quadratic relationship.
- 课：7.1 A Different Kind of Change；7.2 How Does It Change?
- 海报要点：平方项表示二次关系；判断图案是线性、指数还是二次
- 海报：`a1-u7a-另一种变化`「这是哪种变化？」

B部分 二次函数（Section B: Quadratic Functions）

- 学习目标：Interpret quadratic functions that represent a physical phenomenon, given expressions and graphs.
- 学习目标：Use graphs, tables, and calculations to show that exponential functions eventually overtake quadratic functions.
- 课：7.3 Building Quadratic Functions from Geometric Patterns；7.4 Comparing Quadratic and Exponential Functions；7.5 Building Quadratic Functions to Describe Situations (Part 1)；7.6 Building Quadratic Functions to Describe Situations (Part 2)；7.7 Building Quadratic Functions to Describe Situations (Part 3)
- 海报要点：物体下落、抛射的高度；二次函数与指数函数比较
- 海报：`a1-u7b-二次函数`「物体怎样下落？」
- 本地化 / 安全：下落高度用米，按 $g \approx 9.8\ \text{m/s}^2$ 取近似；不要从高处抛物做实验。

C部分 二次表达式（Section C: Working with Quadratic Expressions）

- 学习目标：Coordinate a quadratic expression given in factored form and the intercepts of its graph.
- 学习目标：Use the distributive property to write equivalent quadratic expressions from factored into standard form.
- 课：7.8 Equivalent Quadratic Expressions；7.9 Standard Form and Factored Form；7.10 Graphs of Functions in Standard and Factored Forms
- 海报要点：因式形式和一般形式互化；零点就是图像与 x 轴交点
- 海报：`a1-u7c-二次表达式`「二次式有几种写法？」

D部分 二次函数图像的特征（Section D: Features of Graphs of Quadratic Functions）

- 学习目标：Explain how a graph is affected by changing parameters in quadratic expressions written in standard, factored, and vertex forms.

- 学习目标：Use an equation in vertex form to identify the maximum or minimum of a quadratic function.
- 课：7.11 Graphing from the Factored Form；7.12 Graphing the Standard Form (Part 1)；7.13 Graphing the Standard Form (Part 2) (Optional)；7.14 Graphs That Represent Situations (Optional)；7.15 Vertex Form；7.16 Graphing from the Vertex Form (Optional)；7.17 Changing the Vertex
- 海报要点：因式式、一般式、顶点式中参数对图像的影响；由顶点式读最值
- 海报：`a1-u7d-二次函数图像的特征`「参数怎样改变抛物线？」

第8单元 一元二次方程（Unit 8: Quadratic Equations）

单元页：Algebra 1 Unit 8

A部分 求未知输入（Section A: Finding Unknown Inputs）

- 学习目标：Write quadratic equations, and reason about their solutions in terms of a situation.
- 课：8.1 Finding Unknown Inputs；8.2 When and Why Do We Write Quadratic Equations?
- 海报要点：写二次方程表示情境，结合情境思考解
- 海报：`a1-u8a-求未知输入`「已知输出求输入？」

B部分 解一元二次方程（Section B: Solving Quadratic Equations）

- 学习目标：Given a quadratic expression of the form ax^2+bx+c, create an equivalent expression in factored form.
- 学习目标：Use factored form and the zero product property to solve quadratic equations.
- 课：8.3 Solving Quadratic Equations by Reasoning；8.4 Solving Quadratic Equations with the Zero Product Property；8.5 How Many Solutions?；8.6 Rewriting Quadratic Expressions in Factored Form (Part 1)；8.7 Rewriting Quadratic Expressions in Factored Form (Part 2)；8.8 Rewriting Quadratic Expressions in Factored Form (Part 3)；8.9 Solving Quadratic Equations by Using Factored Form；8.10 Rewriting Quadratic Expressions in Factored Form (Part 4)
- 海报要点：因式分解；零乘积性质；解的个数
- 海报：`a1-u8b-解一元二次方程`「怎样因式分解求解？」

C部分 配方法（Section C: Completing the Square）

- 学习目标：Comprehend that the “plus-minus” symbol is used to represent both square roots of a number and that the square root notation expresses only the positive square root.
- 学习目标：Solve quadratic equations of the form $x^2+bx+c=0$ by rearranging terms and completing the square.
- 课：8.11 What Are Perfect Squares?；8.12 Completing the Square (Part 1)；8.13 Completing the Square (Part 2)；8.14 Completing the Square (Part 3)；8.15 Quadratic Equations with Irrational Solutions
- 海报要点：± 表示两个平方根；配成完全平方解 $x^2 + bx + c = 0$
- 海报：`a1-u8c-配方法`「怎样配方？」

D部分 求根公式（Section D: The Quadratic Formula）

- 学习目标：Explain whether the products or sums of combinations of rational and irrational numbers are rational or irrational.
- 学习目标：Use the quadratic formula to solve quadratic equations of the form $ax^2+bx+c=0$.

- 课：8.16 The Quadratic Formula；8.17 Applying the Quadratic Formula (Part 1)；8.18 Applying the Quadratic Formula (Part 2) (Optional)；8.19 Deriving the Quadratic Formula；8.20 Rational and Irrational Solutions；8.21 Sums and Products of Rational and Irrational Numbers
- 海报要点：$x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$；有理数与无理数的和与积
- 海报：`a1-u8d-求根公式`「求根公式怎样用？」

E部分 再看顶点式（Section E: Vertex Form Revisited）

- 学习目标：Rewrite a quadratic expression in vertex form to identify the maximum or minimum value of the function that the expression defines.
- 课：8.22 Rewriting Quadratic Expressions in Vertex Form；8.23 Using Quadratic Expressions in Vertex Form to Solve Problems
- 海报要点：配方化成顶点式，求最大值或最小值
- 海报：`a1-u8e-再看顶点式`「最大值在哪里？」

F部分 综合运用（Section F: Let's Put It to Work）

- 说明（官方未列 Section Goals）：
- 课：8.24 Using Quadratic Equations to Model Situations and Solve Problems
- 海报要点：用二次方程建模解决问题
- 海报：`a1-u8f-综合运用`「二次方程怎样建模？」

几何（Geometry）· 8 个单元 · 32 个 Section

第1单元 尺规作图和刚体变换（Unit 1: Constructions and Rigid Transformations）

单元页：Geometry Unit 1

A部分 尺规作图（Section A: Constructions）

- 学习目标：Create a construction from instructions.
- 学习目标：Describe construction steps precisely.
- 学习目标：Use circles in a construction to reason about lengths in figures.
- 课：1.1 Build It；1.2 Constructing Patterns；1.3 Construction Techniques 1: Perpendicular Bisectors；1.4 Construction Techniques 2: Equilateral Triangles；1.5 Construction Techniques 3: Perpendicular Lines and Angle Bisectors；1.6 Construction Techniques 4: Parallel and Perpendicular Lines；1.7 Construction Techniques 5: Squares；1.8 Using Technology for Constructions (Optional)；1.9 Speedy Delivery
- 海报要点：用直尺和圆规作中垂线、等边三角形、角平分线、平行线、正方形
- 海报：`ge-u1a-尺规作图`「只用尺规能作什么？」
- 本地化 / 安全：尺规作图要用圆规，圆规尖端锋利，海报加了安全提示。

B部分 定义刚体变换（Section B: Defining Rigid Transformations）

- 学习目标：Comprehend that rigid transformations produce congruent figures by preserving distance and angles.
- 学习目标：Draw the result of a transformation of a given figure.
- 课：1.10 Rigid Transformations；1.11 Defining Reflections；1.12 Defining Translations；1.13 Incorporating Rotations；1.14 Defining Rotations
- 海报要点：平移、反射、旋转保持距离和角度，得到全等图形
- 海报：`ge-u1b-定义刚体变换`「什么是刚体变换？」

C部分 运用刚体变换（Section C: Working with Rigid Transformations）

- 学习目标：Describe the symmetries of a figure using transformations that take the figure onto itself.
- 学习目标：Explain a sequence of transformations to take a given figure onto another.
- 课：1.15 Symmetry；1.16 More Symmetry；1.17 Working with Rigid Transformations；1.18 Practicing Point-by-Point Transformations (Optional)
- 海报要点：图形的对称；用变换序列把一个图形变到另一个
- 海报：`ge-u1c-运用刚体变换`「怎样用变换序列？」

D部分 证据与证明（Section D: Evidence and Proof）

- 学习目标：Label diagrams in order to write precise conjectures.
- 学习目标：Prove theorems about lines and angles.
- 课：1.19 Evidence, Angles, and Proof；1.20 Transformations, Transversals, and Proof；1.21 One Hundred Eighty
- 海报要点：写准确的猜想；证明对顶角、平行线截线和三角形内角和
- 海报：`ge-u1d-证据与证明`「怎样证明角相等？」

E部分 综合运用：还能作什么（Section E: Let's Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：1.22 Now What Can You Build? (Optional)
- 海报要点：用尺规作图设计图案
- 海报：`ge-u1e-综合运用还能作什么`「还能作出什么图？」

第2单元 全等（Unit 2: Congruence）

单元页：Geometry Unit 2

A部分 全等图形（Section A: Congruent Figures）

- 学习目标：Comprehend and generate congruence statements that establish corresponding parts.
- 学习目标：Justify whether or not figures are congruent by reasoning about rigid transformations.
- 课：2.1 Congruent Parts, Part 1；2.2 Congruent Parts, Part 2；2.3 Congruent Triangles, Part 1；2.4 Congruent Triangles, Part 2；2.5 Points, Segments, and Zigzags
- 海报要点：全等对应：对应边、对应角；用刚体变换判断全等
- 海报：`ge-u2a-全等图形`「什么是全等？」

B部分 三角形全等定理（Section B: Triangle Congruence Theorems）

- 学习目标：Identify situations where Side-Side-Side, Angle-Side-Angle, and Side-Angle-Side Triangle Congruence Theorems apply.
- 学习目标：Use previously proven theorems about triangles to write new proofs.
- 课：2.6 Side-Angle-Side Triangle Congruence；2.7 Angle-Side-Angle Triangle Congruence；2.8 The Perpendicular Bisector Theorem；2.9 Side-Side-Side Triangle Congruence；2.10 Practicing Proofs；2.11 Side-Side-Angle (Sometimes) Congruence (Optional)
- 海报要点：边角边、角边角、边边边；中垂线定理；写证明
- 海报：`ge-u2b-三角形全等定理`「三角形何时全等？」

C部分 四边形的证明（Section C: Proofs about Quadrilaterals）

- 学习目标：Critique others' justifications about quadrilaterals and constructions.
- 学习目标：Use previously proven theorems to write new proofs about quadrilaterals.
- 课：2.12 Proofs about Quadrilaterals；2.13 Proofs about Parallelograms；2.14 Bisect It
- 海报要点：平行四边形的性质证明；评价他人的论证
- 海报：`ge-u2c-四边形的证明`「怎样证明四边形性质？」

D部分 综合运用：四边形全等（Section D: Let's Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：2.15 Congruence for Quadrilaterals
- 海报要点：四边形需要哪些条件才能确定全等
- 海报：`ge-u2d-综合运用四边形全等`「四边形怎样才全等？」

第3单元 相似（Unit 3: Similarity）

单元页：Geometry Unit 3

A部分 位似的性质（Section A: Properties of Dilations）

- 学习目标：Comprehend that dilations take angles to congruent angles and lines not passing through the center of the dilation to parallel lines.
- 学习目标：Create and interpret scaled drawings of figures.
- 课：3.1 Scale Drawings；3.2 Scale of the Solar System (Optional)；3.3 Measuring Dilations；3.4 Dilating Lines and Angles；3.5 Splitting Triangle Sides with Dilation (Part 1)
- 海报要点：位似保角；不过中心的直线变成平行线；比例图
- 海报：`ge-u3a-位似的性质`「位似有什么性质？」

B部分 相似变换与比例推理（Section B: Similarity Transformations and Proportional Reasoning）

- 学习目标：Generate similarity statements about similar figures, and identify equivalent proportional relationships among side lengths.
- 学习目标：Use the Angle-Angle Triangle Similarity Theorem and the Triangle Angle Sum Theorem to show that two triangles are similar.
- 课：3.6 Connecting Similarity and Transformations；3.7 Reasoning about Similarity with Transformations；3.8 Are They All Similar?；3.9 Conditions for Triangle Similarity；3.10 Other Conditions for Triangle Similarity (Optional)；3.11 Splitting Triangle Sides with Dilation (Part 2)；3.12 Practice with Proportional Relationships (Optional)
- 海报要点：相似 = 刚体变换 + 位似；角角相似；三角形中的平行线分线段成比例
- 海报：`ge-u3b-相似变换与比例推理`「三角形何时相似？」

C部分 直角三角形中的相似（Section C: Similarity in Right Triangles）

- 学习目标：Use similarity, proportionality, and the Pythagorean Theorem to solve problems involving right triangles.
- 课：3.13 Using the Pythagorean Theorem and Similarity；3.14 Proving the Pythagorean Theorem；3.15 Converse of the Pythagorean Theorem；3.16 Finding All the Unknown Values in Triangles
- 海报要点：用相似证明勾股定理及其逆定理；求未知边
- 海报：`ge-u3c-直角三角形中的相似`「相似怎样证勾股定理？」

D部分 综合运用：镜子测高（Section D: Let’s Put It to Work）

- 说明（官方未列 Section Goals）：
- 课：3.17 Reflection Similarity
- 海报要点：用反射和相似三角形测量高度
- 海报：`ge-u3d-综合运用镜子测高`「镜子怎样测高度？」
- 本地化 / 安全：用镜子测高时把镜子平放在地面，不要用镜子反射阳光照人眼睛。

第4单元 直角三角形三角学（Unit 4: Right Triangle Trigonometry）

单元页：Geometry Unit 4

A部分 角度与陡度（Section A: Angles and Steepness）

- 学习目标：Comprehend that one acute angle of a right triangle determines all the ratios of the side lengths in that triangle.
- 学习目标：Justify an estimated side length or angle measure using a table of ratios of side lengths of right triangles.
- 课：4.1 Angles and Steepness；4.2 Half a Square (Optional)；4.3 Half an Equilateral Triangle (Optional)；4.4 Ratios in Right Triangles；4.5 Working with Ratios in Right Triangles
- 海报要点：一个锐角决定直角三角形各边之比；比值表
- 海报：`ge-u4a-角度与陡度` 「角决定边的比吗？」

B部分 三角比（Section B: Defining Trigonometric Ratios）

- 学习目标：Calculate angle measures in right triangles using arccosine, arcsine, and arctangent.
- 学习目标：Calculate side lengths in right triangles using cosine, sine, and tangent.
- 学习目标：Use trigonometry to solve problems in context.
- 课：4.6 Working with Trigonometric Ratios；4.7 Applying Ratios in Right Triangles；4.8 Sine and Cosine in the Same Right Triangle；4.9 Trigonometry Squared；4.10 Using Trigonometric Ratios to Find Angles
- 海报要点：sin、cos、tan；由角求边，由边求角（反三角）
- 海报：`ge-u4b-三角比` 「什么是三角比？」

C部分 综合运用（Section C: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：
- 课：4.11 Solving Problems with Trigonometry；4.12 Approximating Pi
- 海报要点：用三角学解决测量问题；近似计算 π
- 海报：`ge-u4c-综合运用` 「三角学能解决什么？」
- 本地化 / 安全：测量情境用米，数据是示例；测高处时不要攀爬。

第5单元 立体几何（Unit 5: Solid Geometry）

单元页：Geometry Unit 5

A部分 旋转体、截面与面积缩放（Section A: Cross-Sections, Scaling, and Area）

- 学习目标：Describe the relationships between scale factors and areas, using square root graphs and calculations.
- 学习目标：Generate multiple cross-sections of three-dimensional figures.
- 学习目标：Identify the three-dimensional solid created by rotating a two-dimensional figure, using a linear axis.
- 课：5.1 Solids of Rotation；5.2 Slicing Solids；5.3 Creating Cross-Sections by Dilating；5.4 Scaling and Area；5.5 Scaling and Unscaling
- 海报要点：旋转得到立体；截面；面积随比例因子的平方变化
- 海报：`ge-u5a-旋转体截面与面积缩放` 「截面和面积怎样变？」

B部分 立体的缩放（Section B: Scaling Solids）

- 学习目标：Comprehend that when a solid is dilated by a scale factor of k, its surface area is multiplied by k^2 and its volume is multiplied by k^3.

- 学习目标：Describe the relationships between scale factors and volumes using cube root graphs and calculations.
- 课：5.6 Scaling Solids；5.7 The Root of the Problem；5.8 Speaking of Scaling
- 海报要点：表面积乘 k^2，体积乘 k^3；立方根
- 海报：`ge-u5b-立体的缩放`「立体放大会怎样？」

C部分 棱柱和圆柱的体积（Section C: Prism and Cylinder Volumes）

- 学习目标：Calculate volumes of solids composed of right and oblique prisms and cylinders.
- 学习目标：Explain connections between ways of finding volumes of prisms and cylinders.
- 课：5.9 Cylinder Volumes；5.10 Cross-Sections and Volume；5.11 Prisms Practice
- 海报要点：V = 底面积 × 高；卡瓦列里原理，斜棱柱同样适用
- 海报：`ge-u5c-棱柱和圆柱的体积`「斜棱柱体积怎么算？」

D部分 棱锥的体积（Section D: Understanding Pyramid Volumes）

- 学习目标：Calculate volumes and dimensions of prisms, cylinders, cones, and pyramids.
- 学习目标：Use decomposition and Cavalieri’ s Principle to informally justify the volume formula V=1/3Bh for pyramids.
- 课：5.12 Prisms and Pyramids；5.13 Building a Volume Formula for a Pyramid；5.14 Working with Pyramids；5.15 Putting All the Solids Together
- 海报要点：V = ⅓Bh；分割和卡瓦列里原理
- 海报：`ge-u5d-棱锥的体积`「棱锥体积怎么算？」

E部分 综合运用（Section E: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：
- 课：5.16 Surface Area and Volume；5.17 Volume and Density；5.18 Volume and Graphing
- 海报要点：表面积与体积；密度 = 质量 ÷ 体积；体积的函数图像
- 海报：`ge-u5e-综合运用`「体积还能用在哪？」
- 本地化 / 安全：密度用克每立方厘米，数字是示例。

第6单元 坐标几何（Unit 6: Coordinate Geometry）

单元页：Geometry Unit 6

A部分 平面中的变换（Section A: Transformations in the Plane）

- 学习目标：Compare and contrast rigid transformations, similarity transformations, and those that are neither.
- 学习目标：Describe transformations as functions that take points in the plane as inputs and give other points as outputs.
- 课：6.1 Rigid Transformations in a Plane；6.2 Transformations as Functions；6.3 Types of Transformations
- 海报要点：变换是输入点、输出点的函数；刚体、相似和其他变换
- 海报：`ge-u6a-平面中的变换`「变换也是函数吗？」

B部分 距离、圆和抛物线（Section B: Distances, Circles, and Parabolas）

- 学习目标：Calculate the center and radius of a circle by completing the square.

- 学习目标：Use the Pythagorean Theorem to write an equation for a parabola given its focus and directrix.
- 课：6.4 Distances and Circles；6.5 Squares and Circles；6.6 Completing the Square；6.7 Distances and Parabolas；6.8 Equations and Graphs
- 海报要点：由勾股定理得圆方程；配方求圆心半径；焦点和准线定义抛物线
- 海报：`ge-u6b-距离圆和抛物线`「圆的方程怎样写？」

C部分 用代数证明几何定理（Section C: Proving Geometric Theorems Algebraically）

- 学习目标：Calculate the coordinates of a point on a line segment that partitions the segment in a given ratio.
- 学习目标：Calculate the intersection point of a triangle’s medians.
- 学习目标：Determine equations for lines, through a given point, that are parallel or perpendicular to a given line.
- 课：6.9 Equations of Lines；6.10 Parallel Lines in the Plane；6.11 Perpendicular Lines in the Plane；6.12 It’s All on the Line；6.13 Intersection Points；6.14 Coordinate Proof；6.15 Weighted Averages；6.16 Weighted Averages in a Triangle；6.17 Lines in Triangles
- 海报要点：平行和垂直直线的斜率；按比例分点；三角形中线交点；坐标证明
- 海报：`ge-u6c-用代数证明几何定理`「坐标怎样证明几何？」

D部分 综合运用（Section D: Let’s Put It to Work）

- 说明（官方未列 Section Goals）：
- 课：6.18 Applying Area and Perimeter on the Plane
- 海报要点：在坐标平面上求面积和周长
- 海报：`ge-u6d-综合运用`「坐标中怎样求面积？」

第7单元 圆（Unit 7: Circles）

单元页：Geometry Unit 7

A部分 线、角和圆（Section A: Lines, Angles, and Circles）

- 学习目标：Determine measures of central angles, inscribed angles, and arcs.
- 学习目标：Use the relationship between radii, chords, and tangent lines to prove geometric theorems.
- 课：7.1 Lines, Angles, and Curves；7.2 Inscribed Angles；7.3 Tangent Lines
- 海报要点：圆心角、圆周角、弧；切线垂直于半径
- 海报：`ge-u7a-线角和圆`「圆中的角有什么规律？」

B部分 多边形与圆（Section B: Polygons and Circles）

- 学习目标：Construct the inscribed and circumscribed circles of a triangle.
- 学习目标：Prove properties of the angles in a quadrilateral inscribed in a circle.
- 课：7.4 Quadrilaterals in Circles；7.5 Triangles in Circles；7.6 A Special Point；7.7 Circles in Triangles
- 海报要点：圆内接四边形对角互补；三角形外接圆和内切圆
- 海报：`ge-u7b-多边形与圆`「三角形的外接圆在哪？」

C部分 圆的度量（Section C: Measuring Circles）

- 学习目标：Comprehend that the radian measure of an angle whose vertex is the center of a circle is the ratio of the length of the arc defined by the angle to the circle’ s radius.
- 学习目标：Coordinate between representations of angles measured in degrees and in radians.
- 学习目标：Generalize methods for calculating lengths of arcs and areas of sectors in circles.
- 课：7.8 Arcs and Sectors；7.9 Part to Whole；7.10 Angles, Arcs, and Radii；7.11 A New Way to Measure Angles；7.12 Radian Sense；7.13 Using Radians
- 海报要点：弧长、扇形面积；弧度 = 弧长 ÷ 半径；角度与弧度互化
- 海报：`ge-u7c-圆的度量`「什么是弧度？」

D部分 综合运用（Section D: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：
- 课：7.14 Putting It All Together
- 海报要点：综合运用圆的知识解决问题
- 海报：`ge-u7d-综合运用`「怎样综合用圆知识？」

第8单元 条件概率（Unit 8: Conditional Probability）

单元页：Geometry Unit 8

A部分 机会（Section A: Up to Chance）

- 学习目标：Use organized lists, tables, and tree diagrams to calculate probabilities for compound events.
- 学习目标：Use the addition rule to calculate probabilities.
- 课：8.1 Up to Chance (Optional)；8.2 Playing with Probability；8.3 Sample Spaces；8.4 Tables of Relative Frequencies；8.5 Combining Events；8.6 The Addition Rule
- 海报要点：样本空间；列表、表格、树状图求复合事件概率；加法公式
- 海报：`ge-u8a-机会`「复合事件概率怎么算？」

B部分 相关事件（Section B: Related Events）

- 学习目标：Determine whether two events are independent or dependent using probability.
- 学习目标：Use the multiplication rule, P(A and B) = P(A | B) ⋅ P(B), to find the probability of an event.
- 课：8.7 Related Events；8.8 Conditional Probability；8.9 Using Tables for Conditional Probability；8.10 Using Probability to Determine Whether Events Are Independent
- 海报要点：条件概率 P(A | B)；乘法公式；判断事件是否独立
- 海报：`ge-u8b-相关事件`「什么是条件概率？」

C部分 综合运用：游戏中的概率（Section C: Let's Put It to Work）

- 说明（官方未列 Section Goals）：
- 课：8.11 Probabilities in Games (Optional)
- 海报要点：分析游戏是否公平
- 海报：`ge-u8c-综合运用游戏中的概率`「这个游戏公平吗？」

代数2（Algebra 2）· 8 个单元 · 33 个 Section

第1单元 数列和函数（Unit 1: Sequences and Functions）

单元页：Algebra 2 Unit 1

A部分 数列（Section A: Sequences）

- 学习目标：Determine missing terms in arithmetic and geometric sequences.
- 学习目标：Determine the growth factor of a geometric sequence and the rate of change of an arithmetic sequence.
- 课：1.1 A Towering Sequence；1.2 Introducing Geometric Sequences；1.3 Different Types of Sequences；1.4 Using Technology to Work with Sequences (Optional)
- 海报要点：等差数列加同一个数，等比数列乘同一个数；补全缺项
- 海报：`a2-u1a-数列`「加还是乘的数列？」

B部分 表示数列（Section B: Representing Sequences）

- 学习目标：Create a table, graph, or recursive definition of a sequence from given information.
- 课：1.5 Sequences Are Functions；1.6 Representing Sequences；1.7 Representing More Sequences
- 海报要点：数列是函数；表格、图像、递推定义
- 海报：`a2-u1b-表示数列`「数列怎样表示？」

C部分 写数列的方程（Section C: What’s the Equation?）

- 学习目标：Create equations for sequences representing situations.
- 学习目标：Identify restrictions on the domain of a sequence based on the context.
- 课：1.8 The nth Term；1.9 What’s the Equation?；1.10 Situations and Sequence Types；1.11 Adding Up
- 海报要点：通项公式；定义域是正整数；等比数列前 n 项和
- 海报：`a2-u1c-写数列的方程`「第 n 项怎样写？」

第2单元 多项式函数（Unit 2: Polynomial Functions）

单元页：Algebra 2 Unit 2

A部分 什么是多项式（Section A: What Is a Polynomial?）

- 学习目标：Generalize that when polynomials are combined by addition, subtraction, or multiplication, the result is a polynomial.
- 学习目标：Identify and interpret relative minimums, relative maximums, the degree, and the constant term of polynomials and their graphs.
- 课：2.1 Let’s Make a Box；2.2 Funding the Future；2.3 Introducing Polynomials；2.4 Combining Polynomials
- 海报要点：做无盖盒子求体积；次数、常数项、相对最大最小值；多项式加减乘
- 海报：`a2-u2a-什么是多项式`「什么是多项式？」
- 本地化 / 安全：做无盖盒子要用剪刀，海报加了安全提示；尺寸用厘米，数字是示例。

B部分 使用多项式（Section B: Working with Polynomials）

- 学习目标：Determine properties of polynomials by representing them in standard or factored form.
- 学习目标：Generate a possible expression for a polynomial function given the horizontal intercepts of the function.
- 学习目标：Identify zeros of polynomial functions written in factored form.
- 课：2.5 Connecting Factors and Zeros；2.6 Different Forms；2.7 Using Factors and Zeros
- 海报要点：一般式与因式式；由零点写多项式
- 海报：`a2-u2b-使用多项式` 「因式和零点什么关系？」

C部分 多项式的图像（Section C: Graphs of Polynomials）

- 学习目标：Calculate the solution to a system of polynomial equations.
- 学习目标：Create equations for polynomial functions with specific end behavior.
- 学习目标：Use zeros and multiplicities to sketch a graph of a polynomial function given in factored form.
- 课：2.8 End Behavior (Part 1)；2.9 End Behavior (Part 2)；2.10 Multiplicity；2.11 Finding Intersections
- 海报要点：端点行为；重数；求两个多项式图像的交点
- 海报：`a2-u2c-多项式的图像` 「多项式图像有何特征？」

D部分 多项式除法（Section D: Polynomial Division）

- 学习目标：Generalize that for a polynomial p(x) and a number a, the remainder on division by (x - a) is p(a).
- 学习目标：Rewrite a polynomial as the product of linear factors.
- 课：2.12 Polynomial Division (Part 1)；2.13 Polynomial Division (Part 2)；2.14 What Do You Know about Polynomials?；2.15 The Remainder Theorem
- 海报要点：长除法；余数定理 p(a)；分解成一次因式
- 海报：`a2-u2d-多项式除法` 「多项式怎样做除法？」

第3单元 有理函数和方程（Unit 3: Rational Functions and Equations）

单元页：Algebra 2 Unit 3

A部分 有理函数（Section A: Rational Functions）

- 学习目标：Identify asymptotes of rational functions from their equations.
- 学习目标：Interpret rational functions in context.
- 课：3.1 Minimizing Surface Area；3.2 Graphs of Rational Functions (Part 1)；3.3 Graphs of Rational Functions (Part 2)；3.4 End Behavior of Rational Functions
- 海报要点：表面积最小的盒子；竖直和水平渐近线
- 海报：`a2-u3a-有理函数` 「有理函数有何特点？」

B部分 有理方程（Section B: Rational Equations）

- 学习目标：Create simple equations involving a rational expression to solve problems.
- 学习目标：Solve equations involving rational expressions, and identify any extraneous solutions.
- 课：3.5 Rational Equations (Part 1)；3.6 Rational Equations (Part 2)；3.7 Solving Rational Equations
- 海报要点：列分式方程解决问题；增根

- 海报：`a2-u3b-有理方程`「分式方程怎样解？」

C部分 多项式恒等式（Section C: Polynomial Identities）

- 学习目标：Prove identities.
- 学习目标：Use the formula for the sum of the first n terms in a geometric sequence to solve problems.
- 课：3.8 Polynomial Identities (Part 1)；3.9 Polynomial Identities (Part 2)；3.10 Summing Up；3.11 Using the Sum
- 海报要点：证明恒等式；等比数列前 n 项和公式
- 海报：`a2-u3c-多项式恒等式`「怎样证明恒等式？」

第4单元 复数和有理指数（Unit 4: Complex Numbers and Rational Exponents）

单元页：Algebra 2 Unit 4

A部分 指数的性质（Section A: Exponent Properties）

- 学习目标：Use both radicals and exponents to represent numbers.
- 课：4.1 Properties of Exponents (Optional)；4.2 Square Roots and Cube Roots (Optional)；4.3 Exponents That Are Unit Fractions；4.4 Positive Rational Exponents；4.5 Negative Rational Exponents
- 海报要点：a^(1/n) 是 n 次方根；有理指数和负有理指数
- 海报：`a2-u4a-指数的性质`「分数指数是什么意思？」

B部分 解含平方根、立方根的方程（Section B: Solving Equations with Square and Cube Roots）

- 学习目标：Calculate solutions to radical equations.
- 学习目标：Explain why a radical equation has 0 solutions or 1 solution.
- 课：4.6 Squares and Square Roots；4.7 Inequivalent Equations?；4.8 Cubes and Cube Roots；4.9 Solving Radical Equations (Optional)
- 海报要点：两边平方可能产生不相等的方程；无解或一个解
- 海报：`a2-u4b-解含平方根立方根的方程`「根式方程怎样解？」

C部分 一种新的数（Section C: A New Kind of Number）

- 学习目标：Add, subtract, and multiply complex numbers, and represent the solutions in the form a+bi.
- 课：4.10 A New Kind of Number；4.11 Introducing the Number i；4.12 Arithmetic with Complex Numbers；4.13 Multiplying Complex Numbers；4.14 More Arithmetic with Complex Numbers (Optional)；4.15 Working Backward
- 海报要点：$i^2 = -1$；复数 a + bi 的加减乘
- 海报：`a2-u4c-一种新的数`「i 是什么数？」

D部分 用复数解二次方程（Section D: Solving Quadratics with Complex Numbers）

- 学习目标：Create quadratic equations that have either real or non-real solutions.
- 学习目标：Solve quadratic equations, and explain the solution method.
- 课：4.16 Solving Quadratics (Optional)；4.17 Completing the Square and Complex Solutions；4.18 The Quadratic Formula and Complex Solutions；4.19 Real and Non-Real Solutions (Optional)
- 海报要点：配方或公式得到非实数解；判别式判断解的类型

- 海报：`a2-u4d-用复数解二次方程`「二次方程无实数解时？」

E部分 综合运用：成比例的圆（Section E: Let's Put It to Work）

- 说明（官方未列 Section Goals）：
- 课：4.20 Drawing Proportional Circles (Optional)
- 海报要点：按面积比例画圆
- 海报：`a2-u4e-综合运用成比例的圆`「怎样画成比例的圆？」

第5单元 指数函数和方程（Unit 5: Exponential Functions and Equations）

单元页：Algebra 2 Unit 5

A部分 增长和衰减（Section A: Growing and Shrinking）

- 学习目标：Interpret the meaning of a rational input to an exponential function in context.
- 学习目标：Understand that an exponential function changes by a constant factor over equal intervals.
- 课：5.1 Growing and Shrinking；5.2 Representations of Growth and Decay (Optional)；5.3 Understanding Rational Inputs；5.4 Representing Functions at Rational Inputs；5.5 Changes Over Rational Intervals；5.6 Writing Equations for Exponential Functions；5.7 Interpreting and Using Exponential Functions (Optional)
- 海报要点：相等间隔乘相同的因子；输入是分数时的意义
- 海报：`a2-u5a-增长和衰减`「分数输入怎么理解？」

B部分 缺失的指数（Section B: Missing Exponents）

- 学习目标：Create equivalent equations in exponential and logarithmic forms.
- 学习目标：Estimate the value of a logarithmic expression.
- 课：5.8 Unknown Exponents；5.9 What Is a Logarithm?；5.10 Interpreting and Writing Logarithmic Equations；5.11 Evaluating Logarithmic Expressions
- 海报要点：对数是指数：$10^x = 1000 \Leftrightarrow x = \log 1000$；估算对数
- 海报：`a2-u5b-缺失的指数`「对数是什么？」

C部分 常数 e（Section C: The Constant e）

- 学习目标：Interpret the parameters in the equations of exponential functions with the form f(t)=P · e^(rt).
- 课：5.12 The Number e；5.13 Exponential Functions with Base e；5.14 Solving Exponential Equations
- 海报要点：f(t) = P · e^(rt)；连续复利
- 海报：`a2-u5c-常数e`「e 是怎样的数？」
- 本地化 / 安全：复利金额用人民币，利率是示例。

D部分 对数函数及其图像（Section D: Logarithmic Functions and Graphs）

- 学习目标：Use graphs and logarithms to calculate or estimate solutions to exponential equations.
- 课：5.15 Using Graphs and Logarithms to Solve Problems (Part 1)；5.16 Using Graphs and Logarithms to Solve Problems (Part 2)；5.17 Logarithmic Functions
- 海报要点：用图像和对数解指数方程；对数函数图像
- 海报：`a2-u5d-对数函数及其图像`「对数函数是什么样？」

E部分 综合运用：对数的应用（Section E: Let's Put It to Work）

- 说明（官方未列 Section Goals）：
- 课：5.18 Applications of Logarithmic Functions (Optional)
- 海报要点：pH、分贝等对数刻度
- 海报：`a2-u5e-综合运用对数的应用`「对数刻度用在哪？」
- 本地化 / 安全：长期处在高分贝环境会损伤听力，海报加了提示。

第6单元 函数的变换（Unit 6: Transformations of Functions）

单元页：Algebra 2 Unit 6

A部分 平移、反射和对称（Section A: Translations, Reflections, and Symmetry）

- 学习目标：Create equations to represent known translations and reflections of a graph.
- 学习目标：Justify why a function is even, odd, or neither from an equation.
- 学习目标：Use function notation to represent a transformation from one graph to another.
- 课：6.1 Matching Up to Data；6.2 Moving Functions；6.3 More Movement；6.4 Reflecting Functions；6.5 Some Functions Have Symmetry；6.6 Symmetry in Equations；6.7 Expressing Transformations of Functions Algebraically
- 海报要点：f(x) + k、f(x − h)、−f(x)、f(−x)；奇函数和偶函数
- 海报：`a2-u6a-平移反射和对称`「函数图像怎样移动？」

B部分 缩放输出和输入（Section B: Scaling Outputs and Inputs）

- 学习目标：Calculate the scale factor k needed to transform the input or output of a function to fit data.
- 学习目标：Create new functions by adding, subtracting, multiplying, and dividing functions.
- 课：6.8 Scaling the Outputs；6.9 Scaling the Inputs；6.10 Combining Functions
- 海报要点：k · f(x)、f(kx)；函数的加减乘除
- 海报：`a2-u6b-缩放输出和输入`「怎样缩放函数？」

C部分 函数的变换（Section C: Transformations of Functions）

- 学习目标：Create a graph of a function given an original function and a transformation.
- 学习目标：Create an equation for a function that has been transformed from a specific function f and an unknown function f(x).
- 课：6.11 Transforming from an Original Function；6.12 Transformation Effects；6.13 Transforming Parabolas；6.14 Transforming Circles (Optional)
- 海报要点：由原函数和变换画新函数；变换抛物线和圆
- 海报：`a2-u6c-函数的变换`「变换顺序重要吗？」

D部分 综合运用：为数据建模（Section D: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：
- 课：6.15 Making a Model for Data
- 海报要点：通过变换让函数拟合数据
- 海报：`a2-u6d-综合运用为数据建模`「怎样让函数贴合数据？」

第7单元 三角函数（Unit 7: Trigonometric Functions）

单元页：Algebra 2 Unit 7

A部分 单位圆（Section A: The Unit Circle）

- 学习目标：Use sine and cosine to find coordinates of points on a circle, in all four quadrants.
- 学习目标：Use the Pythagorean Identity to determine the value of all three trigonometric ratios, given the value of one to start from and the quadrant of the angle.
- 课：7.1 Moving in Circles；7.2 Revisiting Right Triangles；7.3 The Unit Circle (Part 1)；7.4 The Unit Circle (Part 2)；7.5 The Pythagorean Identity (Part 1)；7.6 The Pythagorean Identity (Part 2)；7.7 Finding Unknown Coordinates on a Circle
- 海报要点：单位圆上点的坐标 (cos θ, sin θ)；四个象限；毕达哥拉斯恒等式
- 海报：`a2-u7a-单位圆`「单位圆上的点在哪？」

B部分 周期函数（Section B: Periodic Functions）

- 学习目标：Compare and contrast the features of the cosine, sine, and tangent functions.
- 学习目标：Interpret graphs of cosine and sine for input values between 0 and 2π, values greater than 2π, and values less than 0.
- 课：7.8 Rising and Falling；7.9 Introduction to Trigonometric Functions；7.10 Beyond 2π；7.11 Extending the Domain of Trigonometric Functions；7.12 Tangent；7.13 Some New Ratios (Optional)
- 海报要点：正弦、余弦、正切的图像；超出 0 到 2π 的角
- 海报：`a2-u7b-周期函数`「三角函数图像怎样？」

C部分 三角函数的变换（Section C: Trigonometry Transformations）

- 学习目标：Create trigonometric functions to model circular motion, given a description.
- 学习目标：Identify the midline, amplitude, and period in trigonometric functions presented graphically and with equations.
- 课：7.14 Amplitude and Midline；7.15 Transforming Trigonometric Functions；7.16 Features of Trigonometric Graphs (Part 1)；7.17 Features of Trigonometric Graphs (Part 2)；7.18 Comparing Transformations；7.19 Modeling Circular Motion
- 海报要点：中线、振幅、周期；为摩天轮等圆周运动建模
- 海报：`a2-u7c-三角函数的变换`「怎样为转动建模？」

D部分 综合运用：不只是圆（Section D: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：
- 课：7.20 Beyond Circles
- 海报要点：用三角函数描述其他周期现象
- 海报：`a2-u7d-综合运用不只是圆`「还有哪些周期现象？」

第8单元 统计推断（Unit 8: Statistical Inferences）

单元页：Algebra 2 Unit 8

A部分 研究类型（Section A: Study Types）

- 学习目标：Understand the purpose of randomness in experimental studies, observational studies, and surveys.

- 课：8.1 Being Skeptical；8.2 Study Types；8.3 Randomness in Groups
- 海报要点：实验、观察研究、调查；随机化的作用
- 海报：`a2-u8a-研究类型`「研究为什么要随机？」

B部分 分布（Section B: Distributions）

- 学习目标：Use a normal distribution model to estimate proportions of data within an interval.
- 课：8.4 Describing Distributions (Optional)；8.5 Normal Distributions；8.6 Areas in Histograms；8.7 Areas under a Normal Curve
- 海报要点：正态分布；用正态曲线下面积估计比例
- 海报：`a2-u8b-分布`「正态分布怎样用？」

C部分 样本各不相同（Section C: Not All Samples Are the Same）

- 学习目标：Use data from samples to determine point estimates of population characteristics and to estimate associated margins of error.
- 课：8.8 Not Always Ideal；8.9 Variability in Samples；8.10 Estimating Proportions from Samples；8.11 Estimating a Population Mean
- 海报要点：点估计和误差范围；样本比例和样本均值
- 海报：`a2-u8c-样本各不相同`「样本能代表总体吗？」

D部分 分析实验数据（Section D: Analyzing Experimental Data）

- 学习目标：Use the results from redistributing data from an experiment into groups to determine whether the original difference in means is significant.
- 课：8.12 Experimenting；8.13 Using Normal Distributions for Experiment Analysis；8.14 Questioning Experimenting
- 海报要点：重新随机分组检验均值差是否显著
- 海报：`a2-u8d-分析实验数据`「差异是偶然的吗？」

E部分 综合运用：心率（Section E: Let's Put It to Work）

- 说明（官方未列 Section Goals）：
- 课：8.15 Heart Rates
- 海报要点：设计实验比较心率
- 海报：`a2-u8e-综合运用心率`「运动后心率变多少？」
- 本地化 / 安全：测心率的运动强度要适中，身体不适立即停止，海报加了安全提示；数据是示例。